CalorieAI: Deep Learning-Based Food Calorie Estimation System

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Abstract

An image of food often includes a range of meals and dishes. pictures of food, including numerous dishes, to identify. In food photographs with several dishes, we must identify each one. The precision of object detection, meanwhile, has increased recently. CNN now has a substantially better aesthetic. In this study, we employ the YOLO v8 algorithm, a quicker object identification technique [1], to identify meals and serving utensils. In this investigation, we use food photo datasets to validate the tests. Additionally, this food detector is utilized to calculate the number of calories in food for pictures of various meals. We use YOLO v8 as a food detector to identify each dish in a food image, and image-based food calorie estimation is used to estimate the food calories of each recognized dish [2]. This is how we calculate food calories from a food shot of several meals. For this project, we collect pictures of various meals as well as the total number of calories in each one. Then, using food images of several servings, we estimate food calories by combining the food detector with image-based food calorie estimations.

keywords: Food calorie estimator, CNN, YOLO v8

I.INTRODUCTION

As health consciousness has increased, several food photo recognition programs for monitoring daily meals have been released. However, the majority of them need assistance, particularly when several meals are included in food photographs. While several applications utilize CNN to categorize food dishes, no apps have employed CNN-based dish recognition to yet. Then, in this work, we use a current CNN-based object identification system to find food plates in a multiple-dish lunch photo. The majority of the study on food picture identification that has been published so far, nevertheless, cannot be used directly for food images of many different meals. But in fact, we frequently encounter situations involving many meals, making it difficult to distinguish between them. The Grabcut, region expanding technique, and DPM have been used in earlier research of multiple dishes detection [3]. The objective of contemporary work on object detection is to increase object detection accuracy. To address this issue, the speed of detection is substantially decreasing in the meanwhile. Although it may decrease the detection's accuracy, we apply the YOLO v8 object detection model in this. *A.*

A. OBJECT DETECTION:

According to whether they have a proposal step for an area of interest, the present CNN-based detection network may be structurally split into a one-stage (One-Stage) method and a two-stage (Two-Stage) algorithm. A set of Two-Stage detectors, including R-CNN (2014)[8], Fast R-CNN (2015) [9], and Faster R-CNN (2017), were sequentially suggested by GIRSHICK et al. The YOLO series has received a lot of attention recently with the development of single-stage detectors due to its excellent accuracy and quick speed. Joseph et al. made the initial suggestion for YOLO [10] in 2016. Following that, individuals kept working on it and progressively created new versions, which resulted in continually improving detection performance. The most recent updated model of the current YOLO series is YOLOv8[13]. The overall process is shown in figure 1.

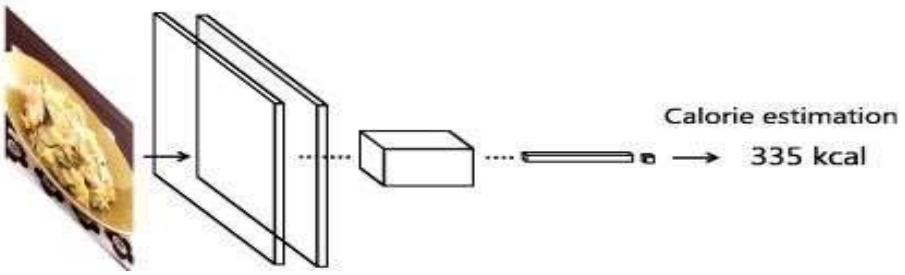

Figure 1. object and calorie prediction

II.RELATED WORKS

A. R-CNN

The task of instance segmentation is carried out by a Mask R-CNN Convolutional Neural Network (CNN) that is based on Deep Learning[1]. The Mask R-CNN generates a mask for the food objects after identifying separate instances of various food things. The mask is then used to calculate the surface area of the identified food item(s). The surface area of the food item and its calorie content per square inch are utilized to calculate its estimated calorie content.

B. Faster R-CNN

According to [5], the faster R-CNN employed in this research is briefly discussed. The foundation of the most recent research on the detection job is faster R-CNN, which has two modules. Region Proposal Network (RPN) is the name of the initial module. The second module, Fast R-CNN detector [6], improves the RPNgenerated region suggestions. The algorithms listed above cannot classify food and perform bounding box regression at the same time, which increases prediction time and renders them useless for real-time objection detection.

III.PROPOSED METHOD

A. YOLO V8:

A rapid and accurate object identification paradigm for computer vision is called YOLO. In contrast to other standard methods, the YOLO method performs object recognition and placement once on the full image rather than splitting it into smaller portions. The YOLO extracts features from pictures using a convolutional neural network (CNN) architecture, and then utilizes convolutional layers to forecast feature positions and kinds. This aids the YOLO's fast processing and precise object identification. The YOLO model has better accuracy and performance than earlier iterations like YOLOv2, YOLOv3, and YOLOv4. The YOLO has been used extensively in real-world settings, including auto-driving, object tracking, object identification, and security surveillance. The head and the backbones are the two fundamental components of the YOLOv8 network design. Convolutional neural networks (CNNs) serve as the framework for the YOLOv8's feature extraction from the analog input photos from the YOLOv5. The EfficientNet or CSPDarknet53 designs, which are potent and effective architectures for feature extraction, are often constructed on top of this convolutional neural network. The CSPLayer used in YOLOv5 has been replaced with the C2f module [12]. The model component positioned at the conclusion of processing makes up the head section in the YOLOv8 architecture. In terms of object identification, classification, and regression, it generates the final predictions. The detection layer, which forecasts the bounding box and the objectness score, and the activation functions, which typically activate the sigmoid function-based objectness scores in YOLOv8, make up the head of the algorithm. These methods display the likelihood that an object is present within a bounding box. The softmax function often turns on categorization probabilities indicating an object's membership in several classes.) Furthermore, YOLOv8 uses impressive loss functions to train the model effectively. Typically, to guarantee that the position prediction of the bounding boxes and the feature classification are carried out reliably and steadily, the loss functions CioU (Complete Intersection over Union) and DFL (Distribution Focal Loss) are utilized [12]. The YOLOv8 successfully recognized and detected objects in photos and videos by combining these loss algorithms.

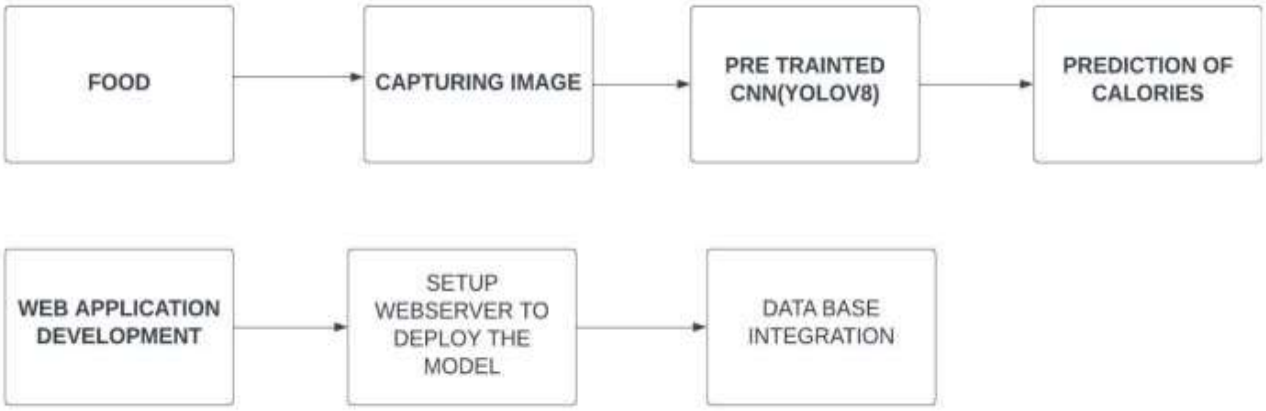

Figure 5. model representation of yolo and real time integration

IV. EXPERIMENT AND RESULTS:

A. DATASETS

The "Food Detection Image Dataset" from the Roboflow platform was used as the dataset for this investigation. Prior to training, preparatory operations like Auto-Orient and uniformly scaling all photos to 256x256 pixels were carried out. After then, the dataset was split into three subsets: a training set, which contained 95% of the data (3376 photos), a validation set, which contained 3% (110 images), and a test set, which contained 2% (57 images). Each training sample was subjected to three variants, including horizontal flips and rotations within a range of -15° to +15°, in order to expand the dataset and improve model resilience. In figure 2 shown as a example of percentage of t. rain, test, validation been used

Figure 2. food dataset 95% train, 3% validation, 2%test data

B. DETECTION OF MULTIPLE DISHES

Yolo v8 has been used to identify the numerous dishes in the cuisine. YOLOv8 is made up of two essential parts: the head and the backbone. Convolutional Neural Networks (CNNs), the foundation, extract information from input pictures. For instance, YOLOv8 makes use of powerful architectures like EfficientNet or CSPDarknet53. Notably, the C2f module has replaced the CSPLayer from YOLOv5. Final predictions are produced by the head section, which is responsible for object detection, categorization, and regression. It also has an activation function that commonly uses sigmoid for objectness scores and softmax for class probabilities, and a detection layer for bounding box and objectness score predictions. In order to provide accurate location prediction of bounding boxes and accurate feature classification, YOLOv8 uses efficient loss functions, CioU and DFL. These techniques aid YOLOv8's successful object detection in both still photos and moving pictures. 3543 photographs, comprising train, test, and validation images, were used to train the model.

During 100 epoches, we trained. As a result, 86 percent of mAP was obtained. Therefore, the different types of food are detected as shown in figure 3.

Figure 3. multiple food detection results

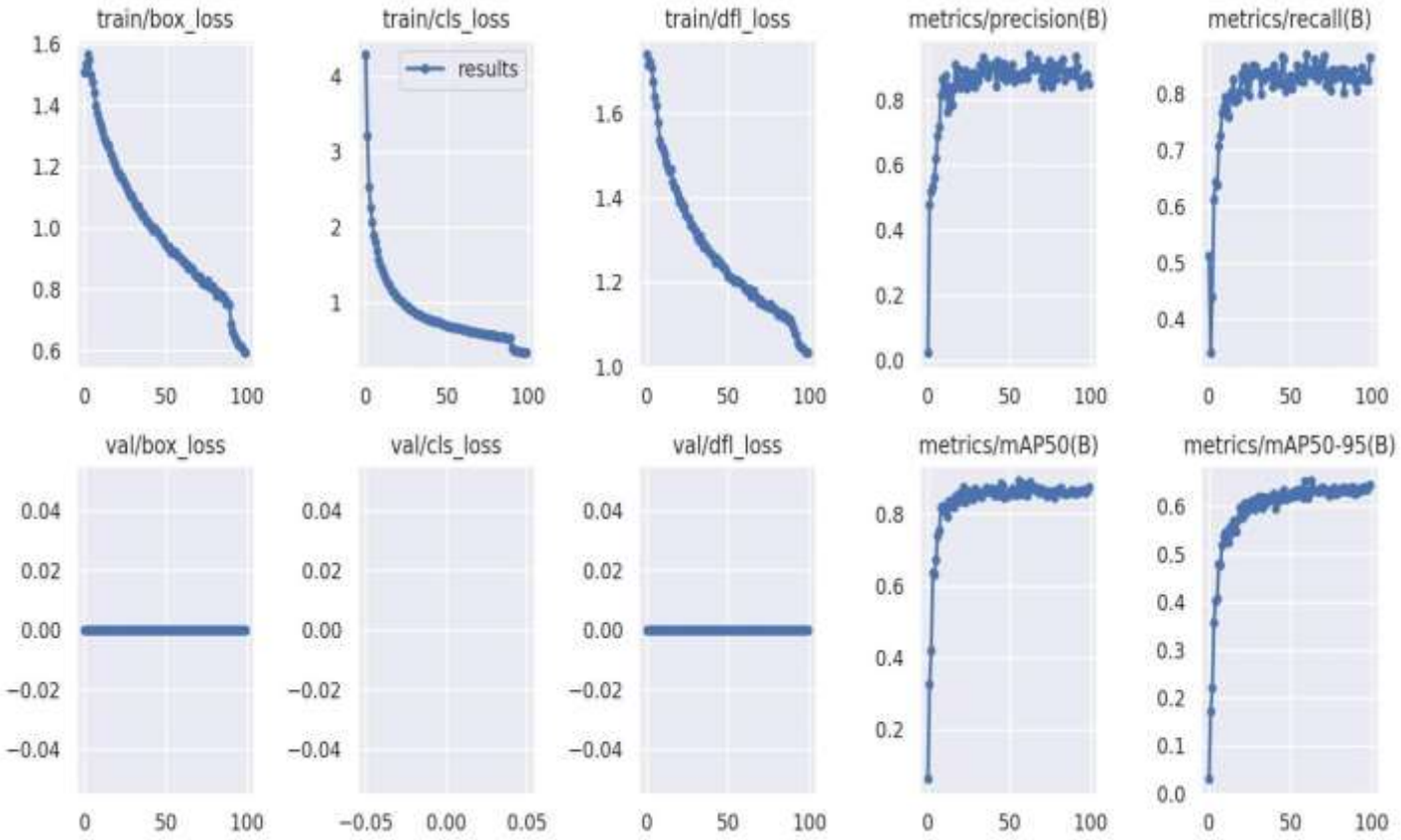

Figure 4. validation loss and mAP curve

C. FOOD CALORIE ESTIMATION OF MULTIPLE DISHES

We've included a technique to estimate the caloric value of certain food components recognized inside a plate to improve the operation of our system. Bounding boxes are created around each identified dish after the successful implementation of YOLOv8 for precise food recognition. We can determine the areas of these bounding boxes using the formula area = width * height by considering the geometric data they offer, including the coordinates of the top-left corner and the dimensions (width and height). We use a carefully curated dataset with a wide variety of foods and their accompanying caloric content to link caloric values to the discovered dishes. The USDA's Food Composition Database is one of the reliable datasets used to create this dataset. By connecting each identified dish to its corresponding record in this dataset, a vital link between visual recognition and dietary data is created. Now that we have established this crucial connection, we can proceed to calculate each dish's calorie composition. This is accomplished by dividing the determined bounding box area by the caloric value of the chosen food according to the dataset. A more thorough evaluation of the total nutritional makeup of the meal may be made thanks to this method, which offers an estimate of the energy content of each individual food item. Additionally, this thorough analysis provides users with specific insights into the calorie content of their meal selections, enabling them to make wellinformed dietary recommendations. This feature also makes our system more adaptable and useful, making it an effective tool for encouraging healthy eating practices and a more active lifestyle. An technique focusing on the identification of each food item's specific area has been developed in order to precisely estimate the caloric content of each food item. This procedure entails taking a picture of the target item from a predetermined distance (D in cm). The recognized food item is then surrounded by a bounding box that may be used to calculate its area. The formulae to calculate the area of object,

$$1/\text{Area in Real Life } (cm^2) = (D/\text{Distance in Real Life } (cm))2$$

-D is the known distance at which the photo was taken from the object (in centimetres).

Each identified food item is connected to the associated nutritional data using a well curated dataset that links dishes with their corresponding caloric values. The determined area of the bounding box is then multiplied by the corresponding caloric value to estimate the amount of calories in each food item.

$$Calories = Abbox \times Cfood$$

$$Total\ Calories = \sum Calories\ food$$

-Abbox represents the area of the bounding box and Cfood denotes the caloric content of the specific food item.

These two expressions are used to calculate the calories in each meal, and then the total number of calories in the entire dish is determined.

V. CONCLUSION

In this study, we used the YOLOv8 object identification algorithm to recognize food items with an impressive 86% accuracy and estimate their caloric value. This important accomplishment highlights the potency of deep learning techniques in the field of food analysis. A solid basis is established for future advancements by the effective integration of YOLOv8 for food identification. By creating a user-friendly website, we want to translate this study into a useful application going ahead. In order to provide customers interested indetermining the caloric content of their meals with a smooth and interactive experience, this platform will make use of Flask, a powerful yet lightweight and adaptable web framework. We intend to enable people to make educated nutritional decisions by using this technology online, encouraging a better way of life. This study emphasizes the scalability of this technology for use in practical applications as well as the promise of YOLOv8 in food analysis. The combination of deep learning and web development has the potential to completely transform how we approach dietary evaluation and planning in light of the rising need for easily available nutritional information. As we go on with the project's next phase, we're excited to grow and improve our system in order to use cutting-edge technology to improve overall health and wellbeing. Overall, this paper underscores the importance of interdisciplinary collaboration between deep learning experts and web developers in addressing contemporary challenges related to nutrition and health. By leveraging cutting-edge technology, such as YOLOv8 and Flask, the research contributes to the ongoing efforts aimed at improving overall health and well-being on a global scale.

VI. REFERENCES

[1] S. Ren, K. He, R. Girshick, and J. Sun. Faster RCNN: Towards realtime object detection with region proposal networks. In Advances in Neural Information
Processing Systems, 2015. 1, 2, 3, 4, 5, 6

[2] T. Ege and K. Yanai. Simultaneous estimation of food categories and calories with multi-task cnn. In Proc. of IAPR International Conference on Machine Vision Applications (MVA), 2017. 1, 3, 5

[3] P. Felzenszwalb, R. Girshick, D. McAllester, and D. Ramanan. Object detection with discriminatively trained part-based models. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2010. 1

[4] Accuracy Improvement of Object Detection in VVC Coded Video Using YOLO-v7 Features Takahiro Shindo, Taiju Watanabe, Kein Yamada, Hiroshi Watanabe.

[5]] R. Girshick, J. Donahue, T. Darrell, and J. Malik. Rich feature hierarchies for accurate object detection and semantic segmentation. In Proc. of IEEE Computer Vision and Pattern Recognition, 2014. 2

[6] R. Girshick. Fast R-CNN. In Proc. of IEEE International Conference on Computer Vision, 2015. 2, 3

[7] YOLOv7-sea: Object Detection of Maritime UAV Images based on Improved YOLOv7 Hangyue Zhao, Hongpu Zhang, Yanyun Zhao

[8[Girshick, Ross. ”Fast r-cnn.” Proceedings of the IEEE international conference on computer vision. 2015.

[9] Ren S, He K, Girshick R, et al. Faster r-cnn: Towards real-time object detection with region proposal networks[J]. Advances in neural information processing systems, 2015, 28.

[10] Redmon, Joseph, et al. ”You only look once: Unified, Realtime object detection.” Proceedings of the IEEE conference on computer vision and pattern recognition. 2016.

[11] Wang, Chien-Yao, Alexey Bochkovskiy, and HongYuan Mark Liao. ”YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors.” arXiv preprint arXiv:2207.02696 (2022).

[12] J. Terven, and D. Margarita Cordova-Esparza, “A Comprehensive Review of Yolo: from Yolov1 and Beyond,” Computer Vision and Pattern Recognition, pp. 1-33, 2023.

[13] YOLO-v1 to YOLO-v8, the Rise of YOLO and Its Complementary Nature toward Digital Manufacturing and Industrial Defect Detection by **Muhammad Hussain** Department of Computer Science, School of Computing and Engineering, University of Huddersfield, Queensgate,
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